

Raunak Kumar Singh

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EDUCATION

University of Illinois Chicago

Master of Science, Computer Science | GPA 3.8/ 4.0

National Institute of Technology, Calicut

Bachelor of Technology, Computer Science

Chicago, Illinois

Jan 2024-Dec 2025

Kozhikode, Kerala

07/2017- 05/2021

SKILLS

Languages: C/C++, Java, JavaScript, Python, TypeScript, Scala, HTML, Tailwind CSS, MySQL, MongoDB, PostgreSQL.

Technologies and Frameworks: Spring Boot, React, Node.js, Next.js, Figma, GraphQL, AWS, GCP, Kafka, Git, Junit, Docker.

TECHNICAL EXPERIENCE

Calamp | Software Engineering Intern

Jun 2025 – Aug 2025

- Developed and enhanced a dynamic fleet management application using Next.js, TypeScript, and React, modernising legacy components and reducing load times by 30% through performance optimisation techniques.
- Implemented new features with improved map rendering for live fleet tracking, leveraging server-side rendering and React Server Components, while increasing Jest test coverage by 40% to improve code reliability.

Deutsche Bank | Senior Software Engineer

July 2021 – Dec 2023

- Architected and implemented a Micro-Frontend architecture for a real-time financial trading platform using Webpack Module Federation, enabling independently deployable modules, optimised web performance, and scalability across the application.
- Architected shared infrastructure for authentication, data sharing, and cross-MFE communication, ensuring seamless coordination across a polyrepo ecosystem while maintaining cross-browser compatibility and mobile responsiveness.
- Implemented robust state management using Redux Toolkit (RTK) and integrated WebSocket-based live data streams to power real-time trading functionality, improving application interactivity, responsiveness, and web performance optimisation.
- Collaborated cross-functionally with UI/UX designers and backend developers in an agile environment to build reusable React components, translating Figma designs into production-ready features that enhanced user experience.
- Streamlined API integration by leveraging RTK Query to consume Spring Boot REST services, improving data consistency through smart caching, reducing API boilerplate, and implementing error handling for complex web interfaces.
- Enforced code quality with ESLint, Jest, and React Testing Library, testing and validating web forms and interactive elements, debugging and troubleshooting web issues, and boosting test coverage across microfrontends.

Cisco | Software Developer Engineer Intern

Jun 2020 – Jul 2020

- Built a collaborative task dashboard using React/Next.js and a secure Node.js backend, integrating role-based access control, drag-and-drop task workflows, JWT authentication, and an optimised, accessible UI.

PROJECTS

[Chicago Rentals](#) [React, Next.js, TypeScript, Responsive Design, Kafka, System Design]

- Built a high-performance rental application using React and Next.js with server-side rendering and static generation, creating responsive UI components with optimised load times, dynamic property search, and streamlined user flows for property discovery and booking.

[Task Lifecycle Management Tool](#) [NestJS, Node.js, TypeScript, Webpack, Jest]

- Designed and implemented a production-grade full-stack workflow system with Next.js and Node.js, combining an accessible drag-and-drop UI, modular backend architecture, JWT authentication, role-based access control, and detailed audit logging.

[COVID TRACKER](#) [REST APIs, OOP, API Integration, Microservices]

- Engineered a Spring Boot backend for a COVID-19 tracker, implementing REST API endpoints and service layers to process, aggregate, and serve real-time country-specific case trends, vaccination statistics, and global health data.

[Hybrid Distributed LLM Architecture](#) [Scala, Hadoop, Apache Spark, gRPC, AWS EMR, Amazon Bedrock, Docker]

- Built a distributed systems project for a cloud computing course that leveraged parallel processing to perform tokenization, vectorisation, and large-scale neural network training using cloud infrastructure.

CERTIFICATION & ACCOMPLISHMENTS

Teaching Assistant (Artificial Intelligence): Served as a Teaching Assistant for an Artificial Intelligence course, holding office hours and supporting the instructor in preparing and evaluating assignments and projects.

Cloud Associate Engineer (Google Cloud Platform): Verified proficiency in cloud infrastructure and services, focusing on deployment and optimization on Google Cloud Platform.